Network Effects of Oil Price Shocks on Inflation

using a DAG-SVAR Framework

Orhun Özel Ece Tașan

Advanced Techniques in Applied Economics

Spring 2026

Motivation: Why Measure Oil Pass-Through?

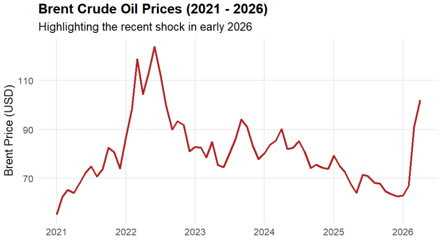
The Macroeconomic Threat

- Oil prices are a primary driver of global inflation dynamics.
- Recent volatility (COVID-19 recovery, Iran conflicts) has renewed central bank focus on energy pass-through.

Motivation: Why Measure Oil Pass-Through?

The Macroeconomic Threat

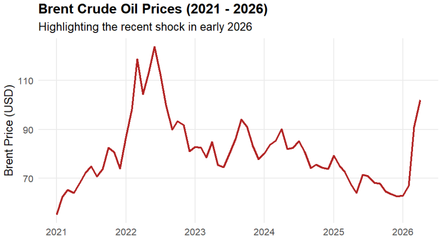
- Oil prices are a primary driver of global inflation dynamics.
- Recent volatility (COVID-19 recovery, Iran conflicts) has renewed central bank focus on energy pass-through.



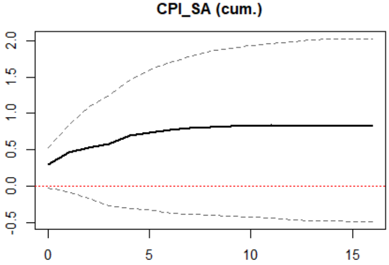
Motivation: Why Measure Oil Pass-Through?

The Macroeconomic Threat

- Oil prices are a primary driver of global inflation dynamics.
- Recent volatility (COVID-19 recovery, Iran conflicts) has renewed central bank focus on energy pass-through.



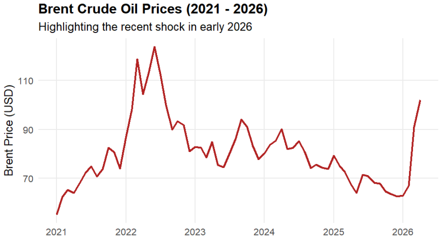
Impulse Response of Inflation to Brent Shock



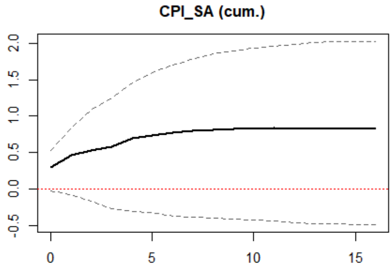
Motivation: Why Measure Oil Pass-Through?

The Macroeconomic Threat

- Oil prices are a primary driver of global inflation dynamics.
- Recent volatility (COVID-19 recovery, Iran conflicts) has renewed central bank focus on energy pass-through.



Impulse Response of Inflation to Brent Shock



The Measurement Problem

Standard time-series methods struggle to isolate this effect accurately due to **high standard errors**.

Related Literature & Our Novelty

Oil Pass-Through & Inflation

- SVAR is the standard tool: Cholesky ordering from oil $\rightarrow$ imports $\rightarrow$ PPI $\rightarrow$ CPI. *(McCarthy, 2006)*
- For Turkey: a 10% Brent shock raises CPI by **0.42 pp** at 1 year *(Akçelik & Öğünç, 2016)*
- Oil shocks affect only a small sub-set of CPI. *(Auburn WP, 2025)*

Causal Discovery in Macroeconomics

- DAG search can replace the arbitrary Cholesky ordering inside SVARs. *(Demiralp & Hoover, 2003)*
- Non-Gaussianity of shocks allows *exact* causal orientation. (LiNGAM / VAR-LiNGAM). *(Shimizu et al., 2006; Hyvärinen et al., 2010; Moneta et al., 2013)*

Related Literature & Our Novelty

Oil Pass-Through & Inflation

- SVAR is the standard tool: Cholesky ordering from oil $\rightarrow$ imports $\rightarrow$ PPI $\rightarrow$ CPI. *(McCarthy, 2006)*
- For Turkey: a 10% Brent shock raises CPI by **0.42 pp** at 1 year *(Akçelik & Öğünç, 2016)*
- Oil shocks affect only a small sub-set of CPI. *(Auburn WP, 2025)*

Causal Discovery in Macroeconomics

- DAG search can replace the arbitrary Cholesky ordering inside SVARs. *(Demiralp & Hoover, 2003)*
- Non-Gaussianity of shocks allows *exact* causal orientation. (LiNGAM / VAR-LiNGAM). *(Shimizu et al., 2006; Hyvärinen et al., 2010; Moneta et al., 2013)*

Our Novelty

Prior work uses DAGs **inside** the SVAR (for ordering). We will use causal discovery to *select* sub-basket and increase the precision of the SVAR.

Methodology: The DAG-SVAR Approach

The Noise Problem

Headline CPI contains irrelevant items (e.g., rent, communication, furniture) that do not react to Brent shocks. This noise dampens the statistical significance of the pass-through in standard VARs.

Methodology: The DAG-SVAR Approach

The Noise Problem

Headline CPI contains irrelevant items (e.g., rent, communication, furniture) that do not react to Brent shocks. This noise dampens the statistical significance of the pass-through in standard VARs.

1. CPI Item Selection (DAG) $\xrightarrow{\text{Filter Noise}}$ 2. Items Aggregation $\xrightarrow{\text{Clean Basket}}$ 3. SVAR

Data & Transformations

Variables (QoQ Changes):

- **CPI & Sub-items** (TurkStat)
- **PPI & Sub-items** (TurkStat)
- **Brent Petroleum Prices**
- **Exchange Rate Basket** (CBRT)
- **Import Prices** (USD, TurkStat)
- **Output Gap** (TurkStat / Estimated)

Seasonality Handling

- Sub-item inflation exhibits intense seasonality.
- We utilize official SA data where available.
- Where unavailable, we applied the **TRAMO-SEATS** algorithm (via Demetra) to extract the seasonally adjusted series before DAG analysis.

DAG with Restricted PC: Network Under Dimensionality Problems

Step 1

Lags of Brent & Motor Fuels $\rightarrow$ First lag of **PPI Sub-items**.

DAG with Restricted PC: Network Under Dimensionality Problems

Step 1

Lags of Brent & Motor Fuels → First lag of **PPI Sub-items**.

→

Step 2

Brent-related items from Step 1 → **PPI Sub-items**.

DAG with Restricted PC: Network Under Dimensionality Problems

Step 1

Lags of Brent & Motor Fuels $\rightarrow$ First lag of **PPI Sub-items**.

$\rightarrow$

Step 2

Brent-related items from Step 1 $\rightarrow$ **PPI Sub-items**.

$\rightarrow$

Step 3

Brent-related items from Step 2 $\rightarrow$ **CPI Sub-items**.

DAG with Restricted PC: Network Under Dimensionality Problems

Step 1

Lags of Brent & Motor Fuels → First lag of **PPI Sub-items**.

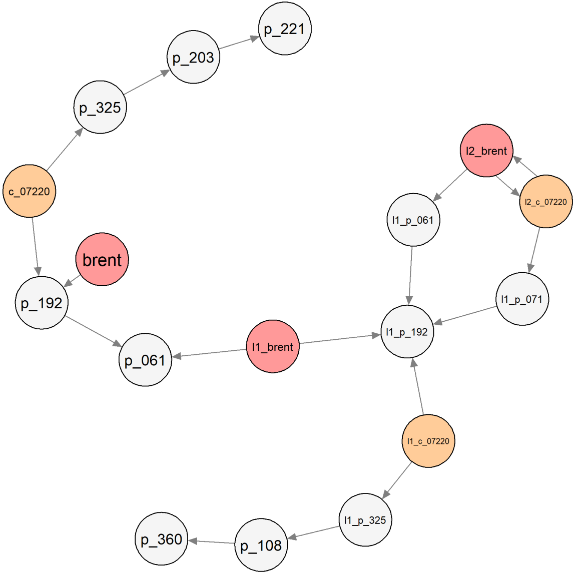
Step 2

Brent-related items from Step 1 → **PPI Sub-items**.

Step 3

Brent-related items from Step 2 → **CPI Sub-items**.

Step 2 Results: The Estimated Propagation Network



Variables

Code	Item Name
c_07220	Fuel for Vehicles (CPI)
p_061	Crude Petroleum
p_192	Refined Petroleum
p_203	Paints & Varnishes
p_221	Rubber Products
p_325	Medical Supplies/Inst.
p_360	Water Treatment
p_071	Iron Ores
p_108	Other Food Products

DAG-Identified Oil Sensitive Inflation Basket

The pc_res Basket

- **Total Items:** 50 CPI Sub-items
- **Total Weight:** 36.47% of the overall CPI basket

Economic Intuition

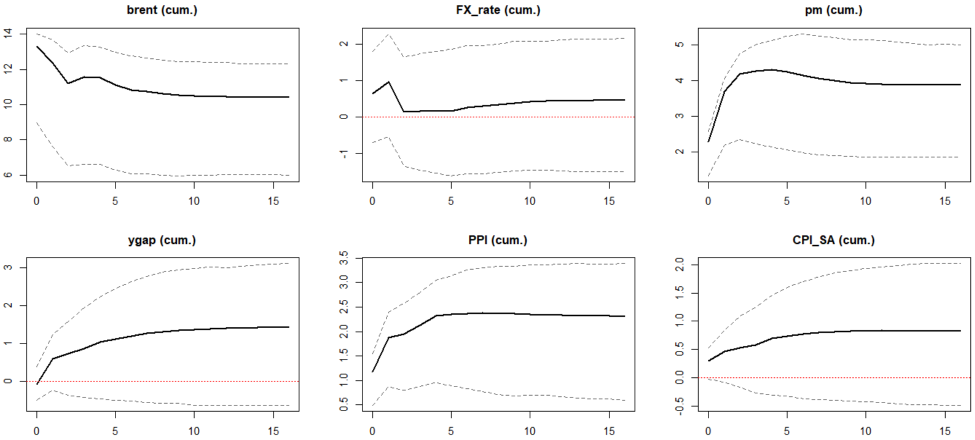
The DAG successfully captures direct energy channels (fuel, natural gas) and highly energy-dependent sectors (transportation, food services, and agricultural products) without human bias.

Top 15 Oil-Sensitive Items (by Weight)

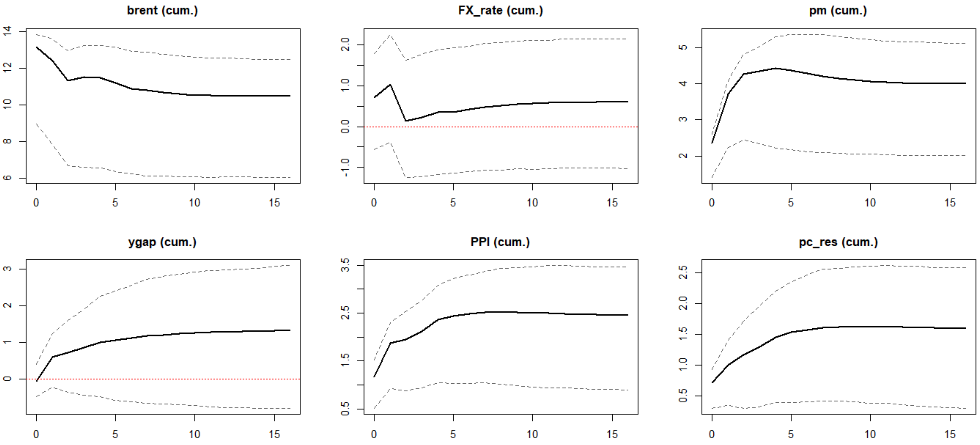
Code	Weight (%)	Item Description
07220	3.75	Vehicle Fuel (Gasoline, Diesel, LPG)
11101	3.66	Restaurants and food services
07113	2.87	New gasoline and electric motor cars
04410	2.67	Water supply
07321	2.49	Passenger transport by local bus
03121	2.20	Garments for men
04522	2.16	Liquefied hydrocarbons
12130	1.50	Personal hygiene, wellness and beauty products
04310	1.43	Materials for maintenance/repair of the dwelling
05611	1.27	Cleaning and maintenance products
01181	1.09	Sugar
01114	1.04	Other bakery products (biscuits, cakes...)
04521	0.95	Natural gas and related subscription fees
01145	0.69	Eggs
07323	0.64	Passenger transport between cities

VAR Impulse Responses: The Benchmark

Model 1: Standard SVAR using Headline `CPI_SA`



Model 2: SVAR using DAG-identified basket (`pc_res`)



Checking the Gaussianity Assumption

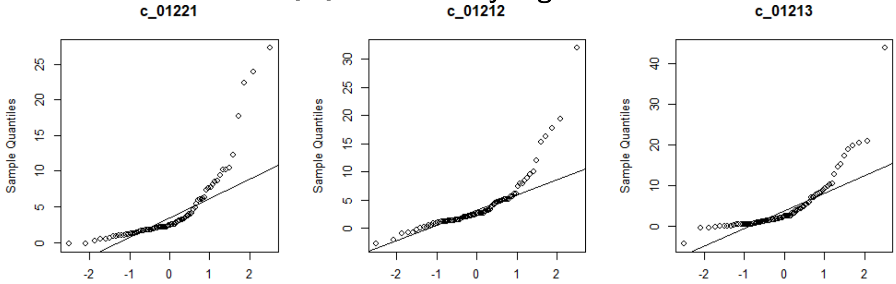
Formal Rejection of Normality

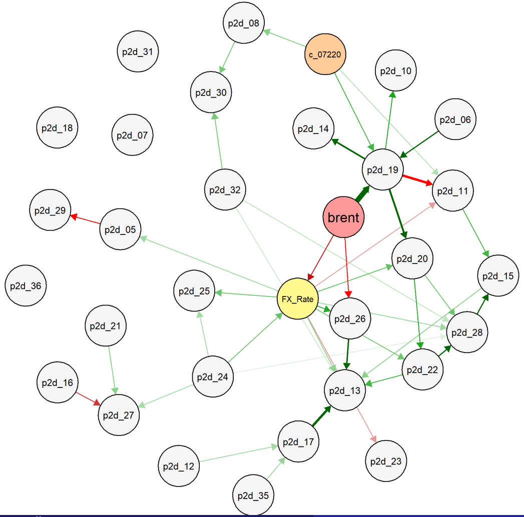
- **Univariate (Anderson-Darling):** Strongly rejects normality for most items.
- **Multivariate (Mardia's Test):** Both Skewness and Kurtosis tests confirm severe joint non-normality ($p < 0.001$).

From Limitation to Resource

Rather than a weakness, non-Gaussianity acts as an **identification resource**. Our next step is **VAR-LiNGAM** *(Shimizu et al., 2006)* which uses non-Gaussian shocks to uniquely orient causal edges and break Markov equivalence.

Q-Q Plots: Heavy Right Tails





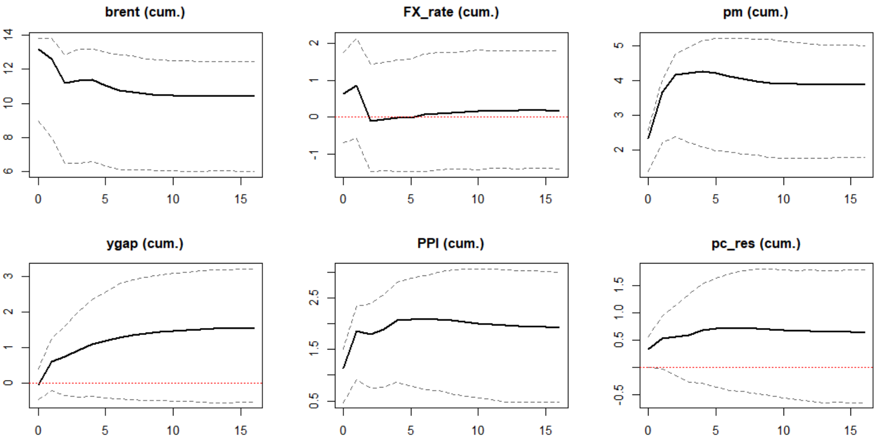
Breaking Markov Equivalence

By exploiting the non-Gaussian residuals, LiNGAM completely identifies the causal direction of all contemporaneous links.

Structural Insights

- **Unambiguous Shocks:** Brent is correctly identified as an exogenous driver strongly impacting **petroleum products**).
- **Full Orientation:** Unlike the standard PC algorithm, there are no remaining undirected edges.

Model 2: SVAR using DAG-identified basket (`pc_res`)



Conclusion

Key Takeaways

1. **Aggregates dilute signals:** Headline CPI averages oil-sensitive and oil-insensitive items, inflating standard errors in standard VARs.
2. **DAGs identify the channel:** The PC algorithm successfully isolated the oil-sensitive sub-basket.
3. **Precision improves:** Replacing headline CPI with DAG identified oil sensitive basket in the SVAR yields significantly tighter confidence bands on the pass-through estimate.
4. **VAR-LiNGAM has costs and benefits:** It can fully orient the causal network using non-Gaussianity, but it has more severe dimensionality problems. It created more volatility in this example.

Appendix: VAR Impulse Responses: The Synthetic Split

Model 3: SVAR with Signal (`pc_res`) + Noise (`pc_res_inv`)